

Lecture n 09

The Endomembrane System

Le Système Endomembranaire

The endomembrane system of eukaryotic cells is comprised of all the cytoplasmic cavities delimited by membranes that communicate with each other via vesicles or canaliculi and also with the plasma membrane. The different compartments of this system are the endoplasmic reticulum, the Golgi apparatus, phagosomes, endosomes, and lysosomes.

1. The endoplasmic reticulum : is a predominant organelle, occupying approximately 20% of the cell volume in eukaryotes. It is isolated from the cytosol by membrane that forms a continuous sheet delimiting an internal space, the "lumen" or "cistern," and creating a network of cavities, vesicles, or canaliculi of varying sizes that communicate with each other.

1.1. Structure and distribution of the endoplasmic reticulum : There are two types of endoplasmic reticulum; The rough endoplasmic reticulum (RER) and the smooth endoplasmic reticulum (SER). The RER and SER can coexist in the same cell and communicate with each other, as is the case in liver cells. Electron microscopy shows continuity between the reticular membrane and the nuclear envelope (Figure 1).

A. The rough endoplasmic reticulum (RER) is a network of flattened sacs that communicate with each other. The ER membrane has a trilamellar structure with a smooth inner reticular face, while the outer cytosolic face is lined with ribosomes. It plays a role in protein synthesis and glycosylation.

B. The smooth endoplasmic reticulum (SER) is formed by a network of small vesicles and tubules. These membranes do not contain ribosomes and play a role in the synthesis of lipids and steroid hormones.

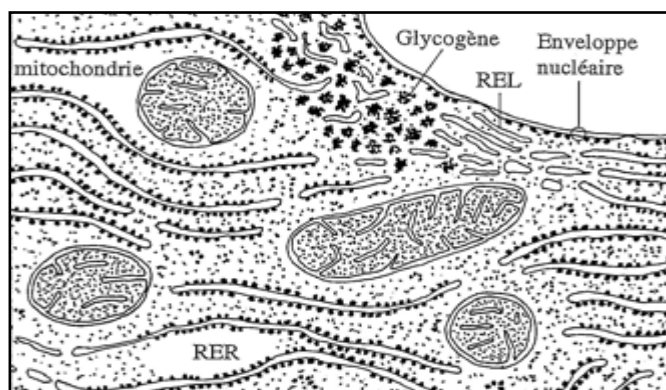


Figure 01: Diagram illustrating the structure and distribution of the smooth and granular endoplasmic reticulum.

1.2. Chemical Composition of the Endoplasmic Reticulum: Biochemical studies performed on the endoplasmic reticulum membrane after isolation by centrifugation show that it contains 70% proteins (generally enzymes) and 30% lipids. The endoplasmic reticulum membrane exhibits a marked asymmetry, as the cytosolic face contains cytochrome b₅, cytochrome P450 reductase, and Ca⁺⁺-dependent ATPase, while the luminal face contains glucose-6-phosphatase, β -glucuronidase, and glycosyltransferases.

1.3. Physiological Roles of the Smooth and Rough Endoplasmic Reticulum: The physiological functions performed by the ER are crucial for cellular life.

1.3.1. Protein Synthesis is the process of protein biosynthesis. It begins in the nucleus, where the transcription phase takes place, following which an mRNA is released into the cytosol. Of all the ribosomes, only those that have begun protein synthesis in the free cytoplasm are directed to the endoplasmic reticulum membranes. From this point on, the fate of the proteins produced by these ribosomes is fixed, and only a limited number of compartments will be accessible to them (Figure 2).

The second step begins with the binding of the mRNA at its 5' end to the free ribosome. This mRNA has a special nucleotide sequence at its 5' end, after the AUG codon, which, once translated, gives rise to the formation of a peptide with a hydrophobic residue called the signal peptide.

An active compound called the signal recognition particle (SRP), composed of six polypeptides linked to an RNA molecule, binds to the N-terminal region of the signal peptide, blocking protein synthesis. The SRP-signal peptide complex then travels to the ER membrane, where it binds to a receptor on the SRP called the anchoring protein. After contact, the SRP dissociates from the complex by hydrolyzing a GTP molecule and is released into the cytosol.

Meanwhile, ER membrane proteins maintain contact with the ribosome (ribosome receptor) via their large subunit, and protein synthesis resumes. A membrane pore formed by a translocation protein opens to allow the protein being synthesized to enter and translocate across the lipid bilayer of the RER. Peptidases cleave the signal peptide from the protein, which is then degraded by proteases in the ER.

Finally, the synthesized protein is released into the lumen of the ER where it acquires its final form.

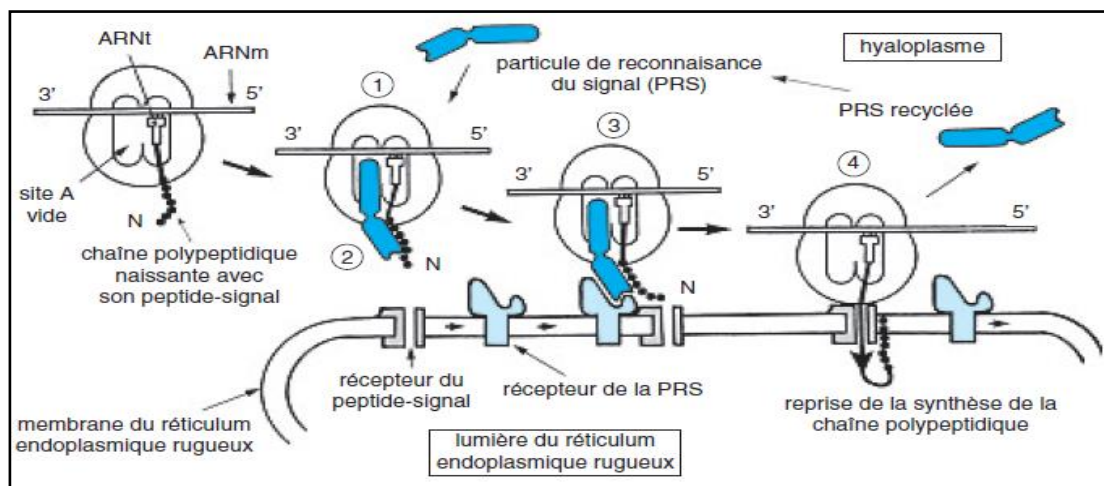


Figure 02: Diagram illustrating the co-translational insertion mechanism of proteins in the rough endoplasmic reticulum.

1.3.2. Protein Glycosylation is the addition of a polysaccharide to a protein or lipid molecule. In the case of proteins, this is a glycoprotein (Figure 3). The steps in the initiation phase of glycosylation in the rough endoplasmic reticulum (RER) begin with the interaction in the cytosol of intermediate nucleotide-carbohydrate residues with dolichol, a doubly phosphorylated membrane isoprenoid.

On the cytoplasmic side, two N-acetylglucosamine (NAG) residues and five mannose residues are added to dolichol via a pyrophosphate bond. The resulting oligosaccharide is then translocated to the luminal side of the ER, where four mannose residues and three glucose residues are added.

A membrane enzyme called oligosaccharide transferase links this oligosaccharide block to the NH₂ group of asparagine found in a consensus sequence (ASN-X-SER/THR) of the protein being synthesized; this glycosylation is called N-glycosylation. Once the glycoprotein is formed, three glucose residues and one mannose residue are removed from the carbohydrate portion. All these reactions constitute the initiation phase of N-glycosylation; further rearrangements will take place in the Golgi apparatus.

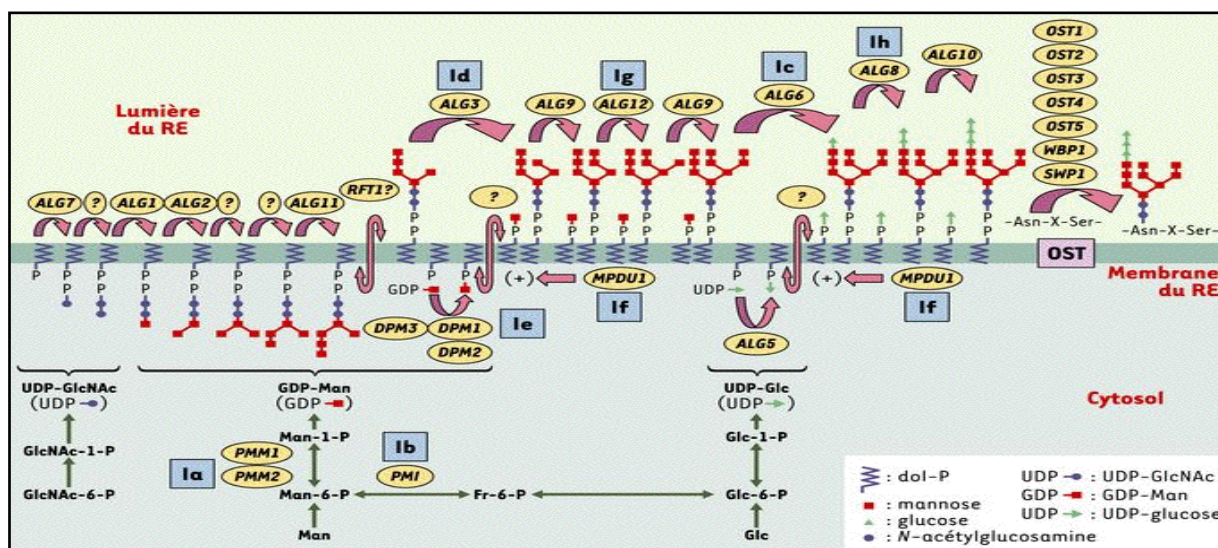


Figure 03. Diagram illustrating the mechanism of N-glycosylation of proteins in the rough endoplasmic reticulum.

1.3.3. Lipid Biosynthesis The smooth endoplasmic reticulum (SER) membrane provides almost all the lipids (phospholipids and cholesterol) necessary for the regeneration of membrane systems. Example of phospholipid synthesis: It begins with glycerol 3-phosphate (G3P) to which fatty acids are added, as in the biosynthesis of phosphatidylcholine described below (Figure 4). Acyltransferases successively add two fatty acids to G3P, yielding phosphatidic acid. A diglyceride is then produced by the action of a phosphatase. In the presence of cytidine diphosphocholine (CDP-choline) and the enzyme choline phosphotransferase, phosphatidylcholine is obtained.

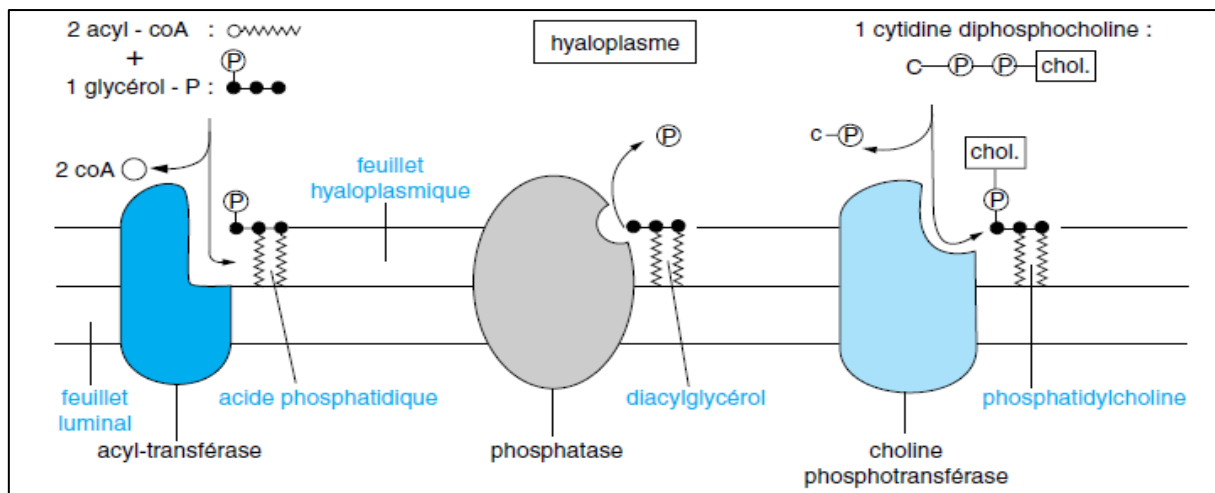


Figure 04. Diagram illustrating the biosynthesis mechanism of phosphatidylcholine.

2. The Golgi Apparatus :

The discovery of this organelle is attributed to the Italian biologist Camillo Golgi (Nobel Prize 1906). It consists of a series of parallel, flattened cisternae or saccules associated with secretory vesicles. A Golgi duct (dictyosome) contains 4 to 6 saccules, and there may be a connection between the lumens of two adjacent saccules. The Golgi apparatus is highly developed in cells specialized in secretion, such as the goblet cells of the intestinal epithelium.

2.1. The Structure of the Golgi Apparatus:

The Golgi apparatus is a polar structure with two faces (Figure 5):

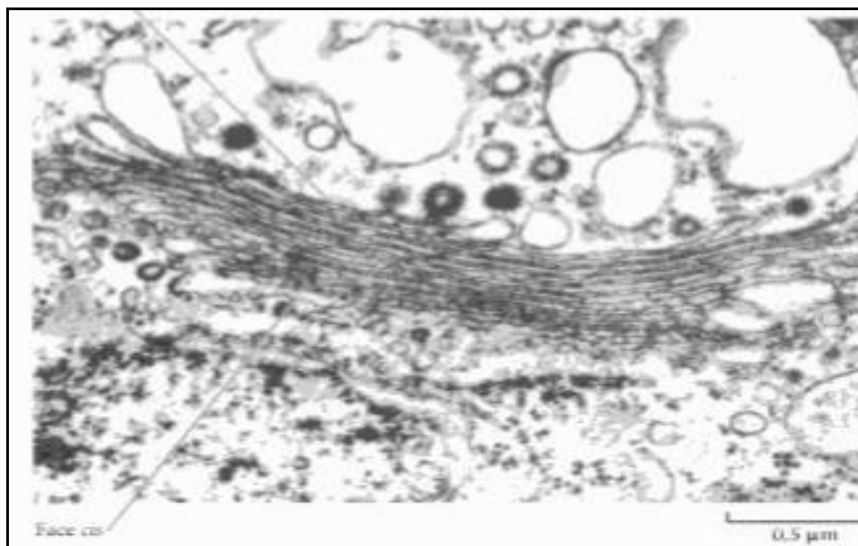


Figure 05. Micrograph showing the structure of a Golgi apparatus.

An outer or cis face is located near the endoplasmic reticulum (ER). At this level, a specialized region of the ER buds off to form transition vesicles. These vesicles pass through the cis Golgi network and fuse with a membrane-bound vesicle depending on their contents.

An inner or trans face is formed of more dilated saccules. At its periphery is a trans Golgi network through which molecules exit, either towards lysosomes, secretory vesicles, or the cell surface (Figure 6).

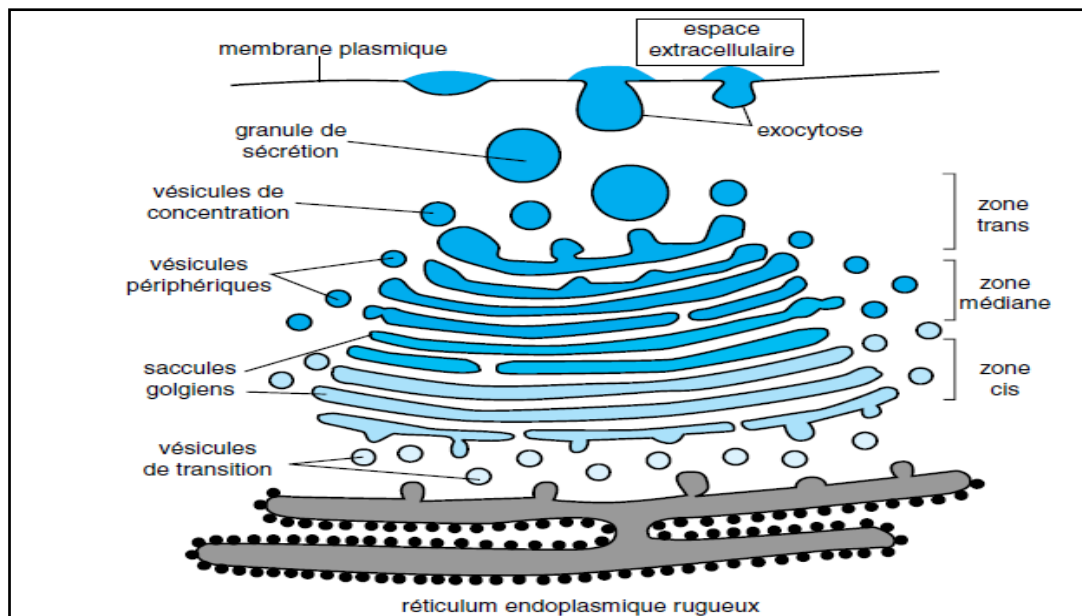


Figure 06. Diagram describing the two faces of the Golgi apparatus.

2.2. Chemical Composition of the Golgi Apparatus

The Golgi apparatus (GA) membranes contain 35 to 40% lipids, primarily phospholipids, and 60 to 65% proteins. The GA is unique in that it contains a significant number of glycosyltransferases, sulfotransferases, and phosphotransferases. The GA exhibits biochemical polarity, as its two faces are biochemically distinct.

2.3. Physiological Roles of the Golgi Apparatus

The main functions attributed to the GA are the continuation of N-glycosylation, O-glycosylation, and the labeling of lysosomal hydrolases.

2.3.1. The N-Glycosylation Sequence

Takes place in the Golgi apparatus, in several steps and successively in the three Golgi compartments, by adding or subtracting different carbohydrate residues from the chain previously formed in the rough endoplasmic reticulum (RER).

2.3.2. O-Glycosylation

Takes place in the Golgi apparatus, more precisely in the medial and trans saccules. In this glycosylation, the glycosylated chain is transferred to the oxygen atom of the serine or threonine amino acid in the protein by an O-glycosyltransferase.

Unlike N-glycosylation, the oligosaccharide chain is highly variable and is not necessarily linked by an N-acetylglucosamine group.

2.3.3. Lysosomal Hydrolase Labeling

This is the phosphorylation or modification essential for the maturation of soluble enzymatic glycoproteins (N-glycosylated) in lysosomes and their targeting to this compartment.

It occurs in the cis-cocci of the Golgi apparatus and takes place in several steps:

- Labeling of mannose molecules by phosphotransferases (by adding a phosphate group to the 6th carbon of one or more mannose molecules).

Once labeled, the enzymes can no longer undergo enzymatic modifications in the rest of the Golgi apparatus.

- Enzymes carrying mannose 6P are recognized by their specific receptors on TGNs.
- Formation of clathrin-coated vesicles that bud from TGNs.
- Removal of the clathrin coating from the vesicle after budding. - Fusion of the vesicles with the endolysosome.

Under the influence of the acidic pH of the endolysosome, the protein detaches from its receptor:

- The enzymes mature after the phosphate group is removed by phosphatases.
- The free mannose 6P receptors are recycled to the Golgi apparatus (clathrin vesicle).

2.3.4. The various pathways followed by proteins:

Proteins synthesized in the ER travel through the Golgi apparatus via transition vesicles (Figure 7). They enter the Golgi apparatus via the cis face (CGN), pass through the median saccules, and reach the trans saccules. They are then packaged into secretory vesicles.

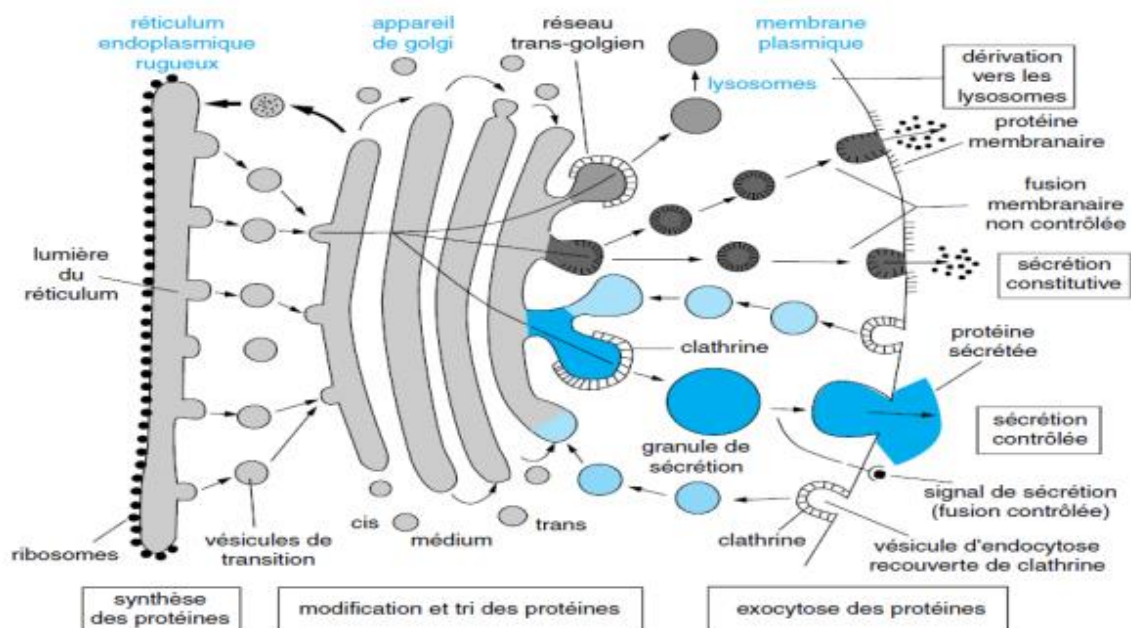


Figure 07: The various pathways followed by proteins.